USER MANUAL

Version 20260904

VERSION 1.0
Issue Date 20260705

WARNINGS/ADVICE

Thank you for choosing KeyMaster. I hope it becomes a useful addition to your station. Please read the following information before connecting the unit.

- **DO NOT** Hot swap the serial connectors. The KeyMaster must be powered down (remove the DC jack) before swapping any serial connector. You have been warned, you may get away with it once, twice or even 10 times, but hot swapping serial connectors has the potential to damage the KeyMaster.
- **DO NOT** exceed the recommended voltage (13.8V)
- Although a 5V supply connector is provided on the rear panel, this is primarily as a convenience when uploaded firmware. The unit has not been tested for long term operation on a direct 5V supply, but there is no reason it will not work.
- **DO NOT** Connect more than one serial interface at any one time – connect to only one radio at a time.
- **DO** Use the same power supply for the KeyMaster as your radio. If you intend to use a different supply, be careful that you do not introduce a different GND - you could cause significant damage by doing this.

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Introduction

KeyMaster is a smart controller for sharing one radio between four CW key inputs. KeyMaster allows up to four different CW keys to remain permanently connected to a single radio. Each key remembers its own CW speed, keyer mode and break-in setting, allowing instant changes between keys without repeatedly adjusting the radio.

KeyMaster has a unique autosense mode, where the KeyMaster senses which key is in use, and automatically switches to route the active key to the radio. The keys can also be switched manually, using buttons on the front panel.

Each of the **four key inputs on KeyMaster has its own saved operating bank of settings**, so changing from one key to another will also recall the chosen CW speed, keyer mode, and break-in mode for that key. Sidetone/monitor volume and band choice is shared across all banks and remains unchanged when switching keys.

The unit can run standalone with no radio connected. When a supported radio is connected by CAT, KeyMaster sends the front-panel changes to the radio. The KeyMaster polls the radio so any changes made directly on the radio interface are read-back by KeyMaster, and the relevant settings updated.

When adjusting WPM, sidetone volume, Keyer ON/OFF and Break-in ON/OFF the KeyMaster displays the current value using the front panel LEDs as a bar chart, or animation.

The KeyMaster features Band select buttons, allowing the operator to quickly switch between bands on the connected radio. This is especially useful for radio such as the Yaesu FTDX10, where there is no direct access to band selection on the front panel.

The KeyMaster **uses an analogue switch, so there is no latency introduced to CW key use**
WIth Key Master auto-keying you can

- Have several keys connected, and a different WPM set on each, allowing super quick changes of speed.
- Have a Bug, Paddle, Straight key and Cootie attached all at the same time, and rapidly switch between them
- Have a bug and paddle connected at the same time, use the paddle to sound a phrase, and then try to sound with the bug. This is a great way to practice bug sending, and get your bug fist as close to perfect as possible.
- Directly adjust WPM, Sidetone volume, Keyer, Break-in, and choose a new band – it’s like having a top-of-the line radio where all the settings are available on the front panel!

Quick Start Guide

1. Connect the KeyMaster to your radio via one of the serial connectors. For example, direct connection via a standard Serial DB9 connector to an FTDX10.
2. Connect the Key output on the rear panel of the KeyMaster to your radio Key input
3. Connect up to four keys to the KeyMaster
4. Connect to the same power supply as your radio (assuming it's in the range 8 to 13.8V) via the DC barrel jack socket.
5. The KeyMaster will start up, and default to the ftDX10 settings. You can select a different radio by (i) Short press Key 3 to activate Key 3 (even if there is no key connected, this is needed to enter the radio setting mode). (ii) Long press Key 3 for about 3 seconds (iii) Choose your radio with the band switches (see manual for which switch corresponds to which radio. Long press Key 3 again to exit band settings
6. Toggle the LED bar graph mode ON/OFF by selecting Key 2, and then long press on Key 2.
7. Toggle auto key switching by selecting Key 1 and then long press on Key 1
8. Now you are up and running, enjoy!

Power

8–13.8V DC

Connections

• KEY OUT -> Radio Key input
• Radio control (12V and 5V Serial)
• Up to four keys

Short Press

Band buttons Select band
Key buttons Select key

Encoder 1

Rotate =increase/decrease WPM
Press = toggle keyer

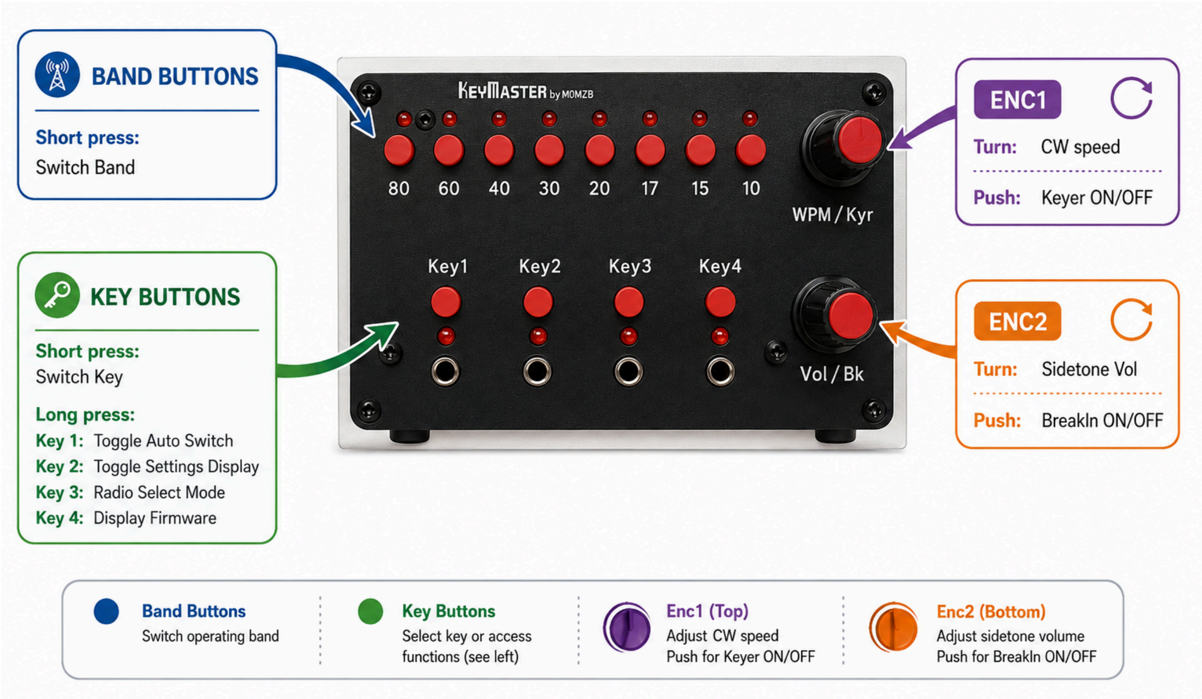
Encoder 2

Rotate = Change Sidetone Volume
Press = toggle break-In

Long Press

Key1 Toggle Auto Switching (auto sensing and switching of keys)
Key2 Toggle LED Display (animated LEDs and bar graph for WPM, volume, etc)
Key3 Radio Select Mode (choose radio and baud rate)
Key4 Display Firmware Version (LEDs display firmware number in binary notation)

Controls Overview



- The KeyMaster includes two rows of buttons.LEDs and two encoders. The top row of buttons allow selection of bands. The LEDs show the chosen band. The Band LEDs are also used as a momentary bar-chart display of settings values, as explained below.
- The Key LEDs/Buttons allow manual selection of the Key to use (manual selection is always available, even if autokey selection is enabled).
- A pair of push-button encoders allow selection of WPM, sidetone volume, Keyer (ON/OFF) and break-in (ON/OFF)
- Four key inputs are provided via 3.5mm sockets. The selected key is routed to the socket on the rear of the unit. An analogue switch is used, so there is no latency introduced.

Band Buttons and Band LEDs

The eight band buttons select:

`80m   60m   40m   30m   20m   17m   15m   10m`

The small LEDs above the band row show the selected band. Some temporary displays reuse the band LEDs for short animations, then return to the selected band display. Where a band-stack register is available on a radio, the KeyMaster will make use of it (i.e. the radio recalls the previous mode and frequency used on that band).

If the settings display is toggled ON (via Key 3 button, see below) the band LEDs show a momentary bar chart of the WPM or volume value when rotating the WPM or Volume encoder. The Band LEDs show a bar graph from left to right, with the value of “9” indicated by alternate LEDs illuminated.
The tens values are shown by the Key LEDs (Key1 LED =10, Key 2=20 etc.).

Key Buttons and Key LEDs

The key buttons can be used to manually select the key in operation. The active key is shown by the corresponding LED. When break-in is off, the active key LED flashes with a short off period. When break-in is on, the active key LED stays solid.

The buttons also have a secondary function that can be accessed by holding for three seconds, but only when that key is chosen. The active key LED blinks after about one second of holding as a warning that the three-second action is about to trigger. To access the secondary function, first short press the key button (if that key is not already chosen) and then long press.

Action	Result
Short press `Key1` to `Key4`	Select that key/input bank
Hold the currently selected `Key1` for 3 seconds	Toggle automatic key-based input switching
Hold the currently selected `Key2` for 3 seconds	Toggle encoder-value display on the LEDs
Hold the currently selected `Key3` for 3 seconds	Enter or leave radio-selection mode
Hold the currently selected `Key4` for 3 seconds	Display firmware version, then send callsign `M0MZB` on the LEDs in Morse

Auto Key Switching

This where the KeyMaster can make your collection of CW keys much more fun! You can have four completely different keys connected to your radio at the same time, and seamlessly switch between them.

With Auto Key Switching Enabled (see above section for instructions on how to do this), the KeyMaster detects which key is in use, and activates that key. A quick press of the left hand paddle (for dual paddles) or closed contact on a straight key is all that is needed for the KeyMaster to select you key.

When the KeyMaster detects a different key being used, relative to the one currently selected. It will firstly disengage all keys, then change the setting on the radio to those stored for the newly selected key (eg turning the Keyer ON/OFF, changing the WPM or changing break-in) and then reactivate the key connection to the radio.

With auto-keying you can

- Have several keys connected, and a different WPM set on each, allowing super quick changes of speed.
- Have a Bug, Paddle, Straight key and Cootie attached all at the same time, and rapidly switch between them
- Have a bug and paddle connected at the same time, use the paddle to sound a phrase, and then try to sound with the bug. This is a great way to practice bug sending, and get your bug fist as close to perfect as possible.

WPM and Keyer Control (Encoder 1)

Action	Result
Turn `WPM/Kyr`	Change CW speed for the active bank
Press `WPM/Kyr`	Toggle keyer mode for the active bank

When encoder-value display is enabled, changing WPM briefly shows the value on the band LEDs before the display returns to normal.

Keyer feedback on the band LEDs: when selecting Keyer ON/OFF the band LEDs provide visual feedback.

Toggling the Keyer OFF
Display shows a single line
(ie a straight key)

Toggling the Keyer ON
Display shows a a pair
Of lines

(ie paddles)

Volume and Breaking Control (Encoder 2)

Action	Result
Turn `Vol/Bk`	Change the shared monitor/sidetone volume
Press `Vol/Bk`	Toggle break-in for the active bank

When encoder-value display is enabled, changing volume briefly shows the value on the LEDs before the display returns to normal.

Break-in feedback on the band LEDs: when selecting Keyer ON/OFF the band LEDs provide visual feedback. When break-in is off, the active key LED flashes with a short off period. When break-in is on, the active key LED stays solid

Settings are saved automatically, but not instantly. Leave the unit powered for at least 30 seconds after changing settings if you want to be certain the latest changes have been written.

Toggling the Break-In OF
The Band LEDs.show a brief animation.
The selected Key LED will blink whenever Break-In is OFF

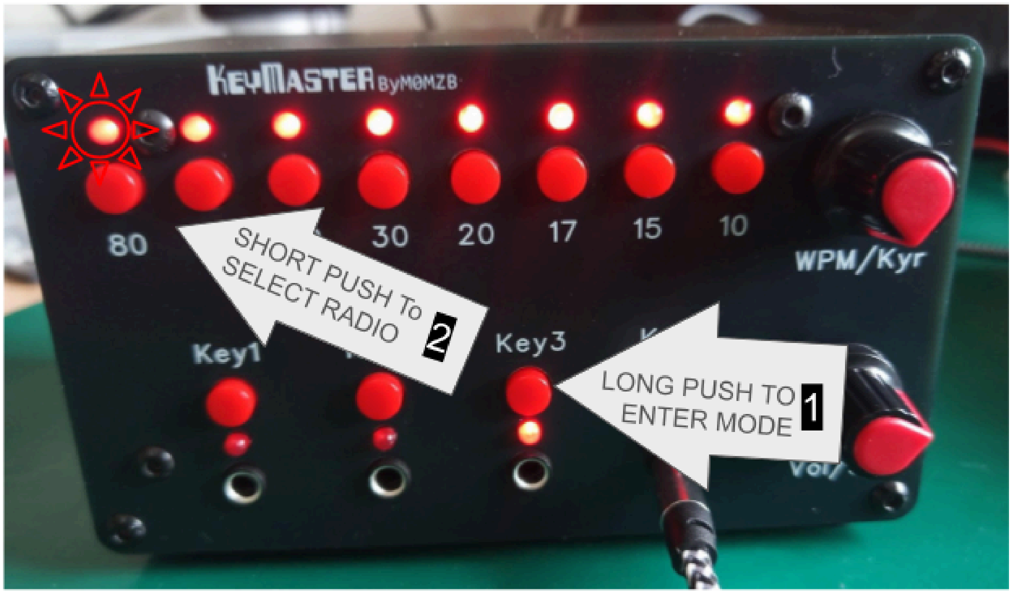
Toggling the Break-In ON
The Band LEDs show the reverse animation.
The selected Key LED is solid whenever Break-In is ON

Adjusting Radio Communication Settings

Selecting the radio

Each radio typically requires a unique set of commands, which have been pre-stored in the KeyMaster firmware. KeyMaster can support up to 255 distinct radios. The following steps allow selection of the radio type:

1. Short press `Key3` so bank 3 is the active bank.
2. Hold `Key3` for 3 seconds to enter radio-selection mode.
3. All band LEDs illuminate. The LED for the current radio flashes.
4. Press one of the band buttons below.
5. Hold `Key3` again for 3 seconds to leave radio-selection mode.



Band button	Radio driver
`80` / button 1	Yaesu FTDX10 Yaesu 991A
`60` / button 2	QRP Labs QMX
`40` / button 3	Kenwood TS-590S
`30` / button 4	Kenwood TS-590SG
`20` / button 5	Yaesu FT-710
`17` / button 6	Kenwood TS890S

The radio choice is saved to EEPROM on the next scheduled settings write.

Supported Radios

Radio	Default CAT rate	Implemented status/control
Yaesu FTDX10 Yaesu FT991A	38400 baud	Band, CW speed, keyer, monitor volume, break-in
Yaesu FT-710	38400 baud	Band, CW speed, keyer, monitor volume, break-in
QRP Labs QMX	9600 baud	Band and CW speed status; band, CW speed, keyer, monitor volume, and practice/break-in control
Kenwood TS-590S	9600 baud	Band, CW speed, keyer, and monitor status; band, CW speed, keyer, monitor, and break-in control
Kenwood TS-590SG	9600 baud	Band, CW speed, keyer, and monitor status; band, CW speed, keyer, monitor, and break-in control
Kenwood TS-890S	9600 baud	Band, CW speed, keyer, monitor, and break-in status/control

Yaesu FT818, and Icom IC7300 support in future versions. Selecting an unsupported type falls back to the FTDX10 driver.

Setting the CAT Baud rate

The baud rate is set while in radio-selection mode.

1. Short press `Key3` so bank 3 is the active bank.
2. Hold `Key3` for 3 seconds to enter radio-selection mode.
3. All key LEDs illuminate. The LED for the current baud rate flashes.
4. Press one of the key buttons, shown below, to choose the baud rate.
5. Hold `Key3` again for 3 seconds to leave radio-selection mode.

Key button	CAT baud rate
`Key1`	4800
`Key2`	9600
`Key3`	19200
`Key4`	38400

Menu Setup Requirements for all radios

It is important to ensure that Serial CAT is activated on your radio, and that the baud rate is set to match the baud rate you select on the KeyMaster.

Yaesu FT991A

On the FT991A perform the following menu selections

- **Menu 028:** `RS232C`
- **Menu 029:** `38400`—and configure KeyMaster to the same rate
- Power-down radio and KeyMaster and then connect the KeyMaster to the Yaesu FT991A to the rear `GPS/CAT` DB9 with a straight-through cable. Do not use a null modem cable.
- Ensure you have power-cycled the radio after changing the serial configuration

Note that menu 31 on the 991A sets CAT rate for the USB connection, so you need to be sure to select menu 029 when making the above changes.

Yaesu FTDX10

On the FTDX10 perform the following menu selections:

- **Operation Setting → General → 232C RATE:** `38400`—and configure KeyMaster to the same rate.
- Connect KeyMaster's `12V SERIAL` DB9 connector to the radio's rear `RS-232C` connector using a straight-through DB9 cable. Do not use a null-modem cable.
- Power down the radio and KeyMaster before connecting the cable.

Be sure to change `232C RATE`, not `CAT RATE`; `CAT RATE` controls the USB CAT connection. The FTDX10 connection uses 8 data bits, no parity and 2 stop bits. [Yaesu FTDX10 documentation]

Yaesu FT-710

On the FT-710 perform the following menu selections:

- **Operation Setting → General → TUN/LIN PORT SELECT:** `CAT-3`
- **Operation Setting → General → CAT-3 RATE:** `38400`—and configure KeyMaster to the same rate.

- **Operation Setting → General → CAT-1 CAT-3 STOP BIT:** `1 bit`
- Connect KeyMaster's `5V SERIAL` connection to the rear `TUNER/LINEAR` mini-DIN connector using the appropriate KeyMaster-to-FT-710 cable.
- Power down the radio and KeyMaster before connecting the cable.

The `TUNER/LINEAR` connector provides 5V TTL serial, not RS-232 voltage levels. Do not connect it to KeyMaster's DB9 `12V SERIAL` connector. Selecting `CAT-3` also means that the connector cannot simultaneously operate an external tuner or linear amplifier. Yaesu FT-710 CAT manual

QRP Labs QMX

On a QMX running recent firmware perform the following menu selections:

- **System Config → Serial 1 on AUX:** `ENABLED`
- **System Config → Serial 1 baud:** `9600`—and configure KeyMaster to the same rate.
- Connect KeyMaster's `5V SERIAL` connection to the QMX `AUX` socket using a suitable stereo 3.5mm cable:
  - QMX tip: QMX transmit, connected to KeyMaster 5V receive
  - QMX ring: QMX receive, connected to KeyMaster 5V transmit
  - QMX sleeve: ground
- Power down the QMX and KeyMaster before connecting the cable.

The AUX serial interface is logic-level serial and **must not be** connected to KeyMaster's DB9 `12V SERIAL` connector. Older QMX firmware may not provide the `Serial 1 on AUX` option, so update the QMX firmware if it is absent. QRP Labs QMX documentation

Kenwood TS-590S

On the TS-590S perform the following menu selection:

- **Menu 61 — COM port baud rate:** `9600`—and configure KeyMaster to the same rate.
- Power down the radio and KeyMaster, then connect KeyMaster's `12V SERIAL` DB9 connector to the radio's rear `COM` connector using a straight-through DB9 cable. Do not use a null-modem cable.
- Power-cycle the radio after changing the COM-port rate.

The connection uses 8 data bits, no parity and 1 stop bit. KeyMaster does not use RTS/CTS hardware flow control. Use Menu 61, not Menu 62, which controls the USB connection. Operate the radio in CW or CW-R mode so that break-in commands are not interpreted as voice VOX commands. Note also the special connection information in the following section. Kenwood TS-590S manual

Kenwood TS-590SG

On the TS-590SG perform the following menu selection:

- **Menu 67 — COM port baud rate:** `9600`—and configure KeyMaster to the same rate.
- Power down the radio and KeyMaster, then connect KeyMaster's `12V SERIAL` DB9 connector to the radio's rear `COM` connector using a straight-through DB9 cable. Do not use a null-modem cable.
- Power-cycle the radio after changing the COM-port rate.

The connection uses 8 data bits, no parity and 1 stop bit. KeyMaster does not use RTS/CTS hardware flow control. Use Menu 67, not Menu 68, which controls the USB connection. Operate the radio in CW or CW-R mode so that break-in commands are not interpreted as voice VOX commands. Kenwood TS-590SG manual

Kenwood TS-890S

On the TS-890S perform the following menu selection:

- **Menu 7-00 — Baud Rate (COM Port):** `9600`—and configure KeyMaster to the same rate.
- Power down the radio and KeyMaster, then connect KeyMaster's `12V SERIAL` DB9 connector to the radio's rear `COM` connector using a straight-through DB9 cable. Do not use a null-modem cable.

The connection uses 8 data bits, no parity and 1 stop bit. KeyMaster does not use RTS/CTS hardware flow control. Use Menu 7-00, not Menu 7-01 or 7-02, which configure the USB virtual COM ports. Kenwood TS-890S manual

Special requirements for Kenwood TS-590S and TS-590SG

```
STRAIGHT KEY OR MECHANICAL BUG -> KEYMASTER INPUT
```

```
  Key or bug contact
       o/ o
       |  |
       |  +-------------------------------- SLEEVE (common/GND)
       |
       +----------------------------------- RING (DAH / dash)

  TIP (DIT / dot) ------------------------- NO CONNECTION
                                             Leave isolated/floating
```

```
KEYMASTER OUTPUT -> TS-590 PADDLE

  KeyMaster TIP    (DIT) ------------------ TS-590 TIP    (DIT)
  KeyMaster RING   (DAH) ------------------ TS-590 RING   (DAH)
  KeyMaster SLEEVE (common) --------------- TS-590 SLEEVE (common)

  This is a normal, fully wired TRS-to-TRS paddle cable.
```

The TS-590 family need care in connected paddles and bugs to the radio, due to hardware limitations on the TS-590S/SG. The diagram above shows how a straight key or bug must be wired in order to work. The text that follows explains the hardware issue, and then goes on to describe how to wire your bug or straight key. Note, paddle keys are unaffected by this - just connect those as usual.

The TS-590S and TS-590SG have separate rear-panel inputs: `KEY` is intended for a straight key or external keyer, while `PADDLE` feeds the radio's internal electronic keyer. KeyMaster must connect to the radio's `PADDLE` socket so it can route ordinary iambic paddles as well as straight keys and bugs.

The TS-590 S/SG is peculiar in that the hardware does not allow disabling the keyer when using the paddle socket. The TS-590 S/SG CAT command set cannot change the `PADDLE` socket into a true straight-key input. KeyMaster works around this radio limitation by using the TS-590 S/SG Bug key function.

Selecting Keyer “on” with the KeyMaster disables Bug mode on the TS-590 S/SG and gives normal electronic/iambic operation via the paddle socket on the TS 590-SG. Keyer “off” enables Bug mode on the TS0590 as a means to permit use of both straight keys and bugs.

The KeyMaster makes use of the fact that during Bug mode, the dash contact of the TS-590 behaves as a straight key. With Bug mode selected on the TS-590, the dit contact still generates automatic dots, while the dah contact becomes manually timed and can behave like a straight-key input.

We therefore want to ensure that straight keys and bugs are wired exclusively to the dah contact, with the dit contact left floating (and the common contact connected as normal).

The main thing the user needs to be concerned with is that there is isolation of the dit contact at the straight-key or bug connection into KeyMaster.

In simple terms, this means that

(i) A stereo cable must be used to connect your bug or straight key to the keymaster.

(ii) The dit contact (the jack tip) of the stereo cable must be left floating (not connected) and the dah contact (the ring of the jack) must be used to connect to your key. The GND contact (sleeve of the jack) is connected to your key as usual

As noted abova, an ordinary paddle connects to a KeyMaster input normally, with dit, dah, and common all wired. KeyMaster routes both paddle contacts through its multiplexer and the fully wired output cable to the TS-590 `PADDLE` socket. This is why the output cable must retain its tip/dit connection.

```
                       Selected KeyMaster input
                                  |
        +-------------------------+-------------------------+
        |                                                   |
 Straight key or bug                                 Iambic paddle
 ring + sleeve only                               tip + ring + sleeve
 tip floating                                             all wired
        |                                                   |
        +-------------------------+-------------------------+
                                  |
                        KeyMaster multiplexer
                                  |
                    normal fully wired TRS cable
                       tip + ring + sleeve
                                  |
                                  v
                        TS-590 PADDLE socket
                                  |
                 +----------------+----------------+
                 |                                 |
       Keyer OFF / Bug ON                 Keyer ON / Bug OFF
       dah manually timed                normal iambic keyer
```

Do not use a mono TS plug for a straight key or bug at a KeyMaster input. Depending on the socket construction, it can short a paddle contact to common. Do not join tip and ring together: in Bug mode, grounding tip asks the radio's internal keyer to generate automatic dots.

This arrangement deliberately uses the TS-590 `PADDLE` socket rather than its separate `KEY` socket, allowing KeyMaster to select either a fully wired paddle or a dah-only straight key/bug without moving the radio plug. It remains an emulation imposed by the TS-590 hardware: keyer-off mode is technically Bug mode, not a true straight-key input mode, and CW message memory is unavailable while Bug mode is enabled.

The diagrams assume the radio's dot/dash reversal setting is OFF: tip is dit and ring is dah. If reversal is enabled, those physical roles exchange, so disable reversal when using this wiring.

Encoder Direction Setting

The direction of each encoder can be reversed independently. The usual setting is for clockwise rotation to increase values, but this can be reversed if desired. The LED bar graph is unaffected by reversing the encoder direction.

Encoder	How to reverse its direction
`WPM/Kyr`	Hold its push-button while powering on
`Vol/Bk`	Hold its push-button while powering on

Keep the button held throughout the startup LED chase. When the confirmation LEDs appear, release it. The outermost key/band LEDs mean that the selected encoder is now reversed; the innermost LEDs mean normal direction. Repeat the same procedure to toggle it back. Each direction setting is saved immediately and remains in effect after power-off.

Default Settings

On blank or invalid EEPROM, KeyMaster starts on input 1 with automatic key switching enabled, encoder-value display disabled, shared monitor volume `20`, and the FTDX10 driver selected.

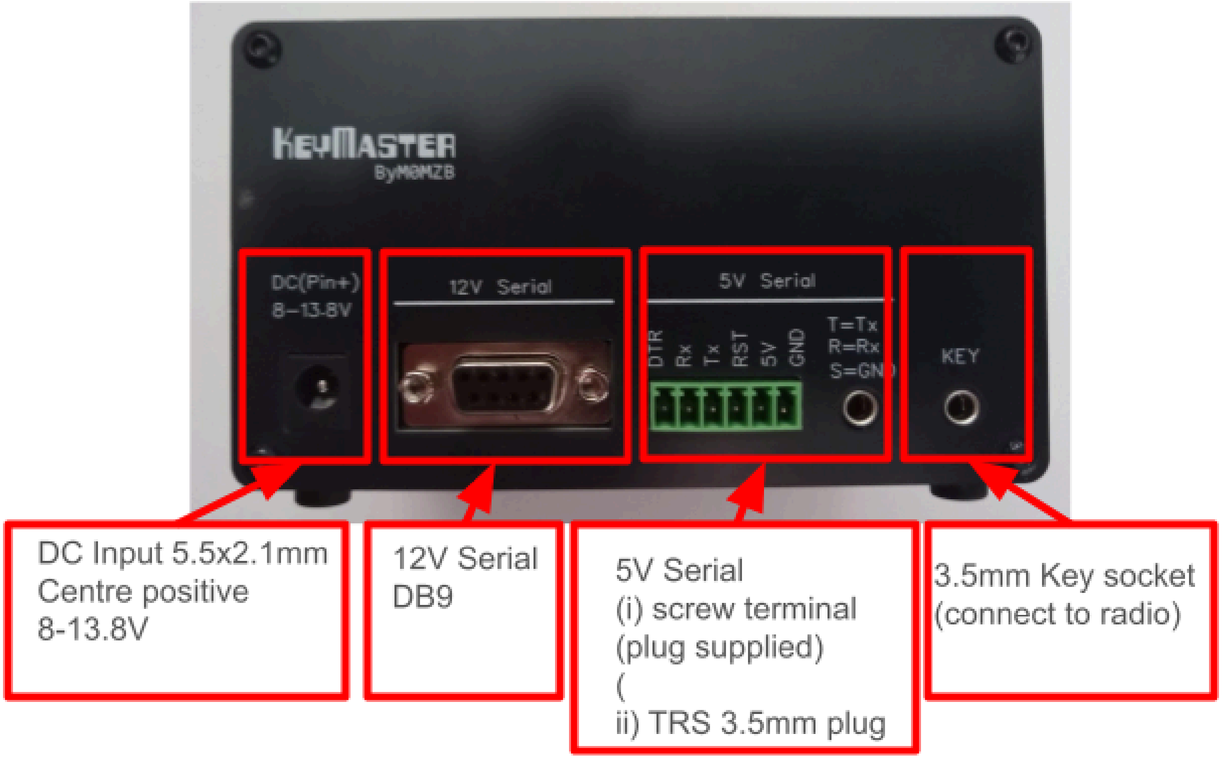
Bank	CW speed	Keyer	Break-in
1	25	On	Off
2	35	On	Off
3	25	Off	Off
4	35	Off	Off

LED Messages Summary

Display	Meaning
One key LED solid	Active key/input bank
Active key LED flashing	Break-in is off for the active bank
One band LED solid	Active band
All band LEDs on, one flashing	Radio-selection mode
Band LEDs show a temporary bar graph	Encoder-value display for WPM or volume
Every other band LED and every key LED lit briefly	CAT command queue overflow warning
Key LEDs flashing during a button hold	Long-press warning

The overflow warning means the firmware could not queue a CAT command at that moment. The unit continues running; if the radio is disconnected or not responding, KeyMaster should still operate as a standalone key/input controller.

Rear Panel Connections



- A standard 5.5x2.1mm DC jack is used to provide power (user supplied)
- A standard DB9 serial cable is used to connect the 12V Serial to a radio. This connection is perfect for the Yaesu FTDX10 which has a DB9 connector on the rear. Just connect the radio and KeyMaster with a standard DB9 Serial cable
- 5V serial is provided, with a screw terminal plug supplied with the KeyMaster to allow easy make-up of your own cables to suit any radio. 5V Serial is also provided via a 3.5mm socket (Tip=Tx, Ring=Rx and Sleeve=GND). It is very easy to make your own cable with readily available 3.5mm plugs.
- The Key output is a 3.5mm socket that connects to the radio Key input

Troubleshooting

Symptom	Things to check
Radio does not respond to controls	Confirm the selected radio type, CAT cable, radio CAT baud rate, and that the radio is powered
KeyMaster works until the radio is unplugged	This should not happen in current firmware; if it does, note the LED pattern and selected radio type
Active key LED keeps flashing	Break-in is off for that bank; press `Vol/Bk` to toggle it
The wrong key bank is selected when keying	Check whether automatic key-based input switching is enabled
A long press does nothing	Make sure you are holding the currently selected key button, not an inactive bank button
Latest setting was lost after power-off	Leave the unit powered for 30 seconds after changing settings
Uploading firmware fails	Disconnect or isolate the radio CAT interface while programming